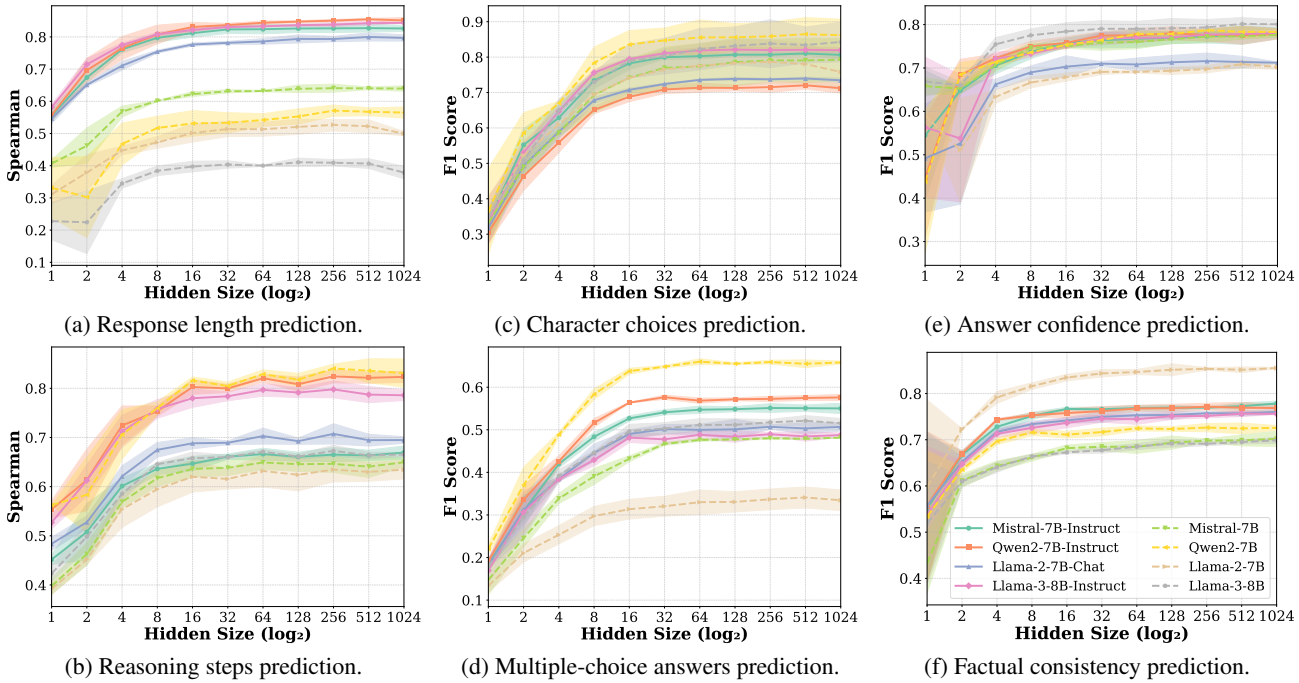


(a) Response length prediction. (c) Character choices prediction. (e) Answer confidence prediction.

(b) Reasoning steps prediction. (d) Multiple-choice answers prediction. (f) Factual consistency prediction.

Figure 4. Hidden-size study results. Performance of MLP probes plateaus at relatively small hidden sizes (≤ 128) across all tasks, with structure attributes converging around size 16, content attributes at 32, and behavior attributes at 8. This suggests a hierarchy of pattern complexity, with behavioral patterns being most accessible and content patterns requiring more sophisticated probes.

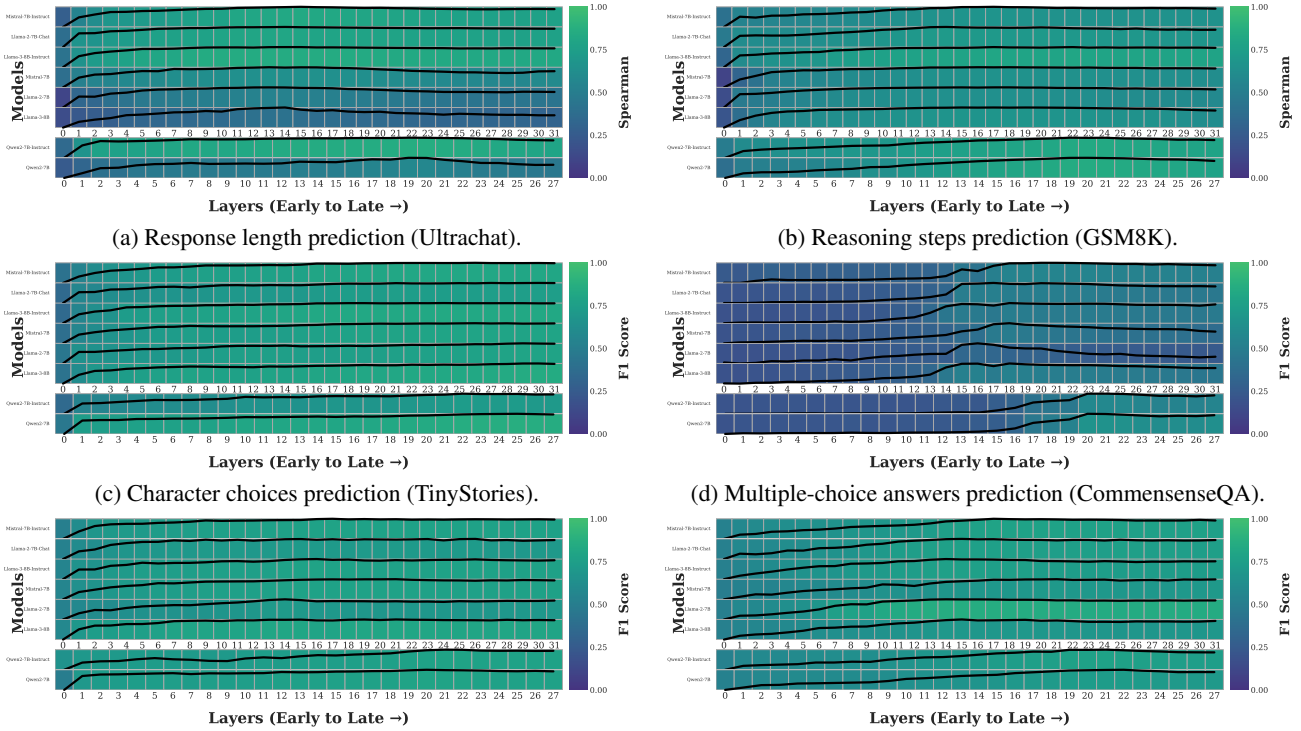
4. Experimental Results

In this section, we present experimental results across six tasks, showing that LLM hidden prompt representations encode rich information about upcoming responses and can be used to probe and predict global response attributes.

Insight 1: Models present emergent planning on structure, content, and behavior attributes, which can be probed with high accuracy (Fig. 2). Our in-dataset probing experiments (where probes are trained and tested on different splits of the same prompt dataset) reveal that both base and fine-tuned models encode structure, content, and behavior attributes, with fine-tuned models showing superior performance. For structural attributes (response length and reasoning steps; Fig. 2a, 2b), fine-tuned models exhibit strong linear correlations with ground truth, clustering around $y = x$ (with example fitting results shown in Fig. 2g, 2h), while base models show weaker but positive correlations. For content and behavior attributes (character choices, multiple-choice answers, answer confidence, and factual consistency; Fig. 2c, 2d, 2e, 2f), both model types demonstrate robust classification performance above random baselines. These findings also suggest that **models develop systematic internal planning representations for content and behavior attributes during pre-training, with structure attributes requiring additional reinforcement through fine-tuning.**

Insight 2: The learned patterns generalize across datasets, indicating intrinsic task-related patterns rather than dataset-specific ones (Fig. 3). Our cross-dataset experiments (training and testing probes on different prompt datasets for the same task, e.g., GSM8K→MATH or TinyStories→ROCStories) demonstrate robust generalization of learned patterns. For structure attributes (Fig. 3a, 3b), predictions maintain strong correlations with target labels despite lower accuracy compared to in-dataset testing (with example fitting results shown in Fig. 3g, 3h), with fine-tuned models showing stronger correlations than base models. Similarly, for content and behavior attributes (Fig. 3c, 3d, 3e, 3f), performance remains above baseline in cross-dataset settings. These results suggest that **probes capture generalizable task-related patterns rather than dataset-specific features, indicating that models may develop intrinsic emergent planning capabilities that transfer across different contexts within the same task domain**.

Insight 3: Emergent planning patterns are salient across models and tasks, extractable with simple MLP probes (Fig. 4). We investigate pattern saliency by varying the hidden size of two-layer MLP probes and measuring their average performance across model layers. Performance plateaus before hidden size 128 across all datasets, with larger sizes that can even lead to overfitting, indicating pattern saliency. The results can also indicate saliency differences across attributes: structure attributes (Fig. 4a, 4b)



(a) Response length prediction (Ultrachat).

(b) Reasoning steps prediction (GSM8K).

(c) Character choices prediction (TinyStories).

(d) Multiple-choice answers prediction (CommensenseQA).

(e) Answer confidence prediction (MedMCQA).

(f) Factual consistency prediction (CREAK).

Figure 5. Layer-wise attribute prediction dynamics. Six subplots (one per task): Y-axis shows eight models; X-axis traces layer-wise progression (early → late). Heatmap colors indicate absolute performance (0–1, lighter = higher); black curves show row-normalized relative capability trends. Key dynamics: Structure attributes peak mid-layers, content attributes exhibit varied emergence timelines but consolidate in later layers, and behavior attributes stabilize early. Layer-wise probing reveals hierarchical organization of planning capabilities, with progressive refinement shaping final outputs.

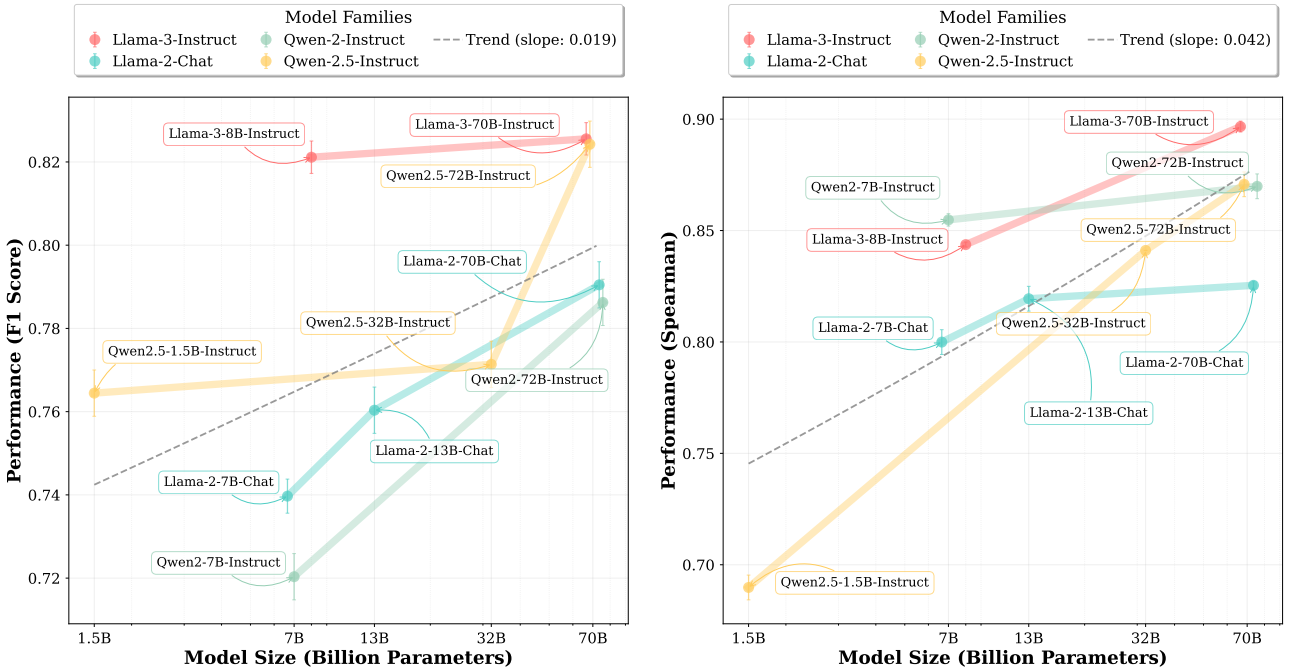
converges around hidden size 16, content attributes (Fig. 4c, 4d) plateau around 32, and behavior attributes (Fig. 4e, 4f) plateaus at around 8, suggesting a hierarchy of representation complexity where behavioral patterns are most readily accessible, structural patterns require moderate complexity to capture, and content patterns demand the most sophisticated probe architectures. The consistent pattern across different model scales and architectures illustrates fundamental organizational principles in language model representations, suggesting that **emergent planning is an inherent property of large language models rather than an artifact of specific architectures or training procedures**.

Insight 4: Attribute patterns accumulate and peak differently across model layers (Fig. 5). We conduct layer-wise probing analysis (with hidden sizes optimized per layer) to understand how different attributes emerge through model layers. The results reveal distinct accumulation patterns for each attribute type. Structure attributes (Fig. 5a, 5b) show weak performance in early layers, peak in middle layers, and partially diminish in final layers, suggesting a gradual accumulation followed by refinement. Content attributes (Fig. 5c, 5d) peak in later layers, either through sudden late-layer emergence or gradual accumulation, indicating their reliance on higher-level semantic processing. Behavior attributes (Fig. 5e, 5f) demonstrate uniform distribution across layers except for the initial few, suggesting they are fundamental properties encoded early in the model. These layer-wise patterns reveal that **(1)** different aspects of planning emerge through distinct computational paths, **(2)** the hierarchical nature of planning, from basic behavioral patterns to complex structural decisions, is reflected in the layer-wise organization, and **(3)** the emergence of these patterns through progressive transformations, rather than from initial embeddings alone, indicates that planning capabilities arise from learned computational processes rather than simple statistical correlations.

5. Ablation

5.1. Planning Ability Scales with Model Size

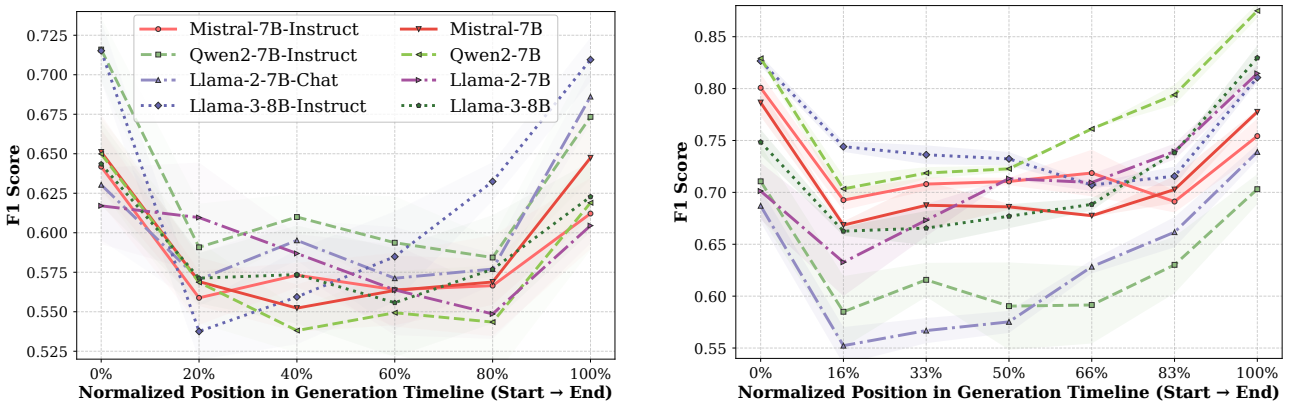
We analyze how emergent response planning scales across different model sizes using four model families: Llama-2-chat (7B, 13B, 70B), Llama-3-Instruct (8B, 70B), Qwen-2-Instruct (7B, 72B), and Qwen-2.5-Instruct (1.5B, 32B, 72B). Using grid search over layers and hidden sizes, we identify



(a) Character choices prediction (TinyStories).

(b) Response length prediction (Ultrachat).

Figure 6. Scaling effects on planning capabilities. Evaluated across four model families (Llama-2-chat, Llama-3-Instruct, Qwen-2-Instruct, Qwen-2.5-Instruct; 1.5B–72B) using UltraChat and TinyStories, structure and content attributes show family-specific scaling: larger models within each family improve planning.



(a) Answer confidence prediction (MedMCQA).

(b) Character choice prediction (TinyStories).

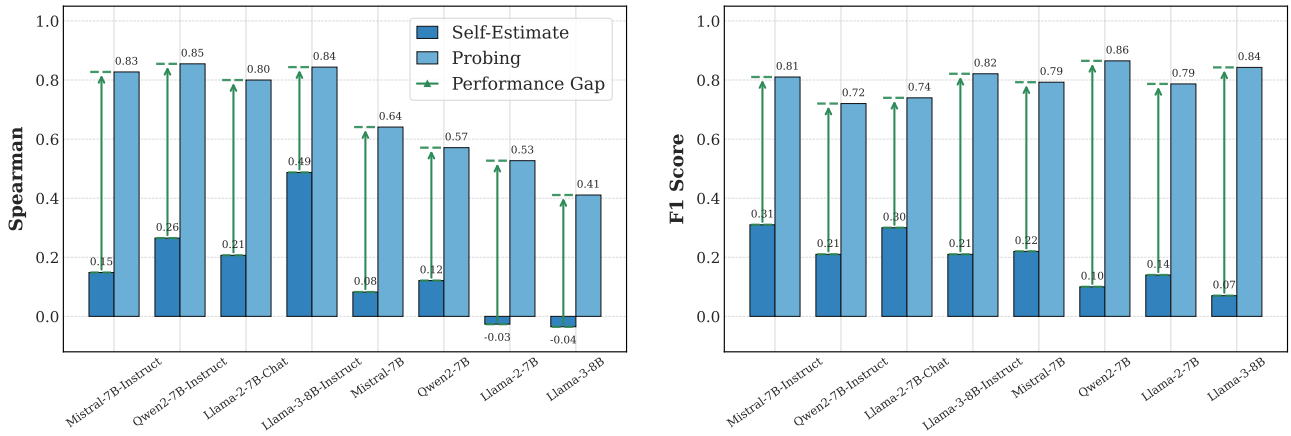
Figure 7. Planning dynamics during generation. Probing at equidistant positions (answer confidence, character choice) shows three-phase patterns: high accuracy in early segments (global planning intent), mid-segment decline (local token focus), and late-stage recovery (contextualized refinement). This suggests models first outline global attributes, then refine locally, before finalizing coherent plans.

optimal configurations and evaluate models on UltraChat and TinyStories datasets, focusing on structure and content attributes. We exclude base models as smaller models have short context which limit few-shot prompts, while the same prompts fail to effectively prompt larger base models to follow instructions. We omit the behavior attribute type as larger models tend to give correct answers consistently, making it difficult to obtain balanced data for analysis.

Fig. 6 exhibit two key insights: **(1)** within each model family, larger models demonstrate stronger planning capabilities, and **(2)** this scaling pattern does not generalize across different model families, suggesting that other factors like architectural differences also influence planning behavior.

5.2. Evolution of Planning Representations During Response Generation

We analyze how planning features evolve during generation by probing at different positions in the response sequence.



(a) Response length prediction.

(b) Character choices prediction.

Figure 8. Gap between probed and verbalized results. Both tasks reveal a systematic gap between verbalized self-estimates (*Self-Estimate*) and probe-based predictions (*Probing*). Base models exhibit near or worse-than-random accuracy in self-estimation, while fine-tuned models achieve only marginal gains. This gap demonstrates that models encode richer planning information in hidden representations than they can explicitly access during generation, revealing a divide between implicit planning and explicit self-awareness.

For each response, we collect activations from the first token up to the token before attribute-revealing keywords (e.g., animal words in story character selection tasks) or throughout the entire sequence for tasks requiring external ground-truth labels (e.g., answer confidence tasks). We divide these positions into equal segments and apply probes previously trained with in-dataset settings at each division point. We conduct experiments on two datasets: TinyStories for character choice prediction and MedMCQA for answer confidence prediction. Fig. 7 reveal a distinctive pattern: **probing accuracy is high initially, decreases in the middle segments, and rises again toward the end**. This pattern suggests a three-phase planning process: **(1)** initial phase with strong planning that provides an overview of the intended response; **(2)** middle phase with weaker planning, characterized by more local, token-by-token generation; **(3)** final phase with increased planning clarity as accumulated context makes the target attributes more apparent.

5.3. Gap Between Probed and Verbalized Results

We investigate whether LLMs can self-estimate their response attributes when explicitly prompted, and compare these verbalized results—self-predictions obtained via direct prompting—against probe-based predictions. Experiments focus on two tasks: response length prediction (Ultrachat) and character choice (TinyStories). For verbalized predictions, we prompt LLMs in separate runs to self-estimate attributes (e.g., "Estimate your answer length in tokens using `[TOKENS]number[/TOKENS]`" for tuned models, or via few-shot examples with pre-calculated lengths for base models). Self-estimation accuracy is evaluated by comparing these outputs against actual greedy-decoded response attributes.

Fig. 8 reveals a systematic gap: base models exhibit near or worse-than-random accuracy in self-estimation, while fine-tuned models improve only marginally, remaining far inferior to probe-based methods. This suggests **models encode more planning information in hidden representations than they can explicitly access during generation**, highlighting a divide between implicit planning and explicit self-awareness.

6. Discussion

6.1. Emergent Response Planning under Sampling

In this study, we consistently use greedy decoding to derive deterministic probing labels $\hat{g}_i = g(\mathbf{y}_i)$ for representations $\mathcal{H}_i = \{\mathbf{H}^l_{\mathbf{x}_i}\}_{l=1}^{L}$. But when generalizing to sampling settings, while greedy decoding simplifies sampling approximation by reflecting the LLM's most probable output, this approach may not fully capture sampling nuances. We propose two potential ways for improvement:

Averaging: Replace greedy labels with attribute averages over multiple sampling trials (e.g., 10 samples) to approximate expected sampling behavior.

Distributional probing: Train probes to predict label distributions instead of single values, capturing uncertainty inherent to sampling. While greedy decoding reflects the LLM's most probable output (approximating sampling averages), distribution-aware probing remains an open challenge, which we leave for future work.